COVID-19 Severity: Predictive Modeling and Risk-Score Analysis

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1. Introduction

The goal is to identify demographic and clinical factors that predict severe COVID-19 illness and to derive a patient-level risk score. From the 10,000-participant study, 1,000 were drawn at random (`set.seed(6107)`) and partitioned 80 / 20 into training (n = 800) and test (n = 200) sets. The outcome is binary (1 = severe, 0 = not severe). All 13 candidate predictors in the codebook were retained — seven continuous (age, height, weight, BMI, SBP, LDL, depression) and six categorical, including the multi-level factors `race` and `smoking`.

2. Exploratory analysis

Severe outcomes occur in 32% of the training sample. There is no missingness and no obvious outliers. Summary statistics by outcome are in **Table 1** and the marginal patterns in **Figure 1**. Three observations drive the modelling choices that follow. **First**, severe patients are older (median age 61 vs 59) and have higher BMI and SBP. **Second**, the empirical severity rate is 62% in unvaccinated vs 10% in vaccinated patients — by far the largest single-variable gap (the next, hypertension, is 41% vs 23%). **Third**, `height`, `weight` and `bmi` are mechanically related (BMI = weight / height^2, with r = 0.70 between weight and BMI and r = −0.50 between height and BMI); `SBP` and `hypertension` correlate at r = 0.80. We kept all 13 codebook predictors and let regularised / tree-based models handle the redundancy.

3. Model training

Nine candidates were trained with `caret::train` on the 800-row training set using **shared 10-fold CV folds** (`trainControl(index = createFolds(...))`) so that every model is compared on identical resamples. Tuning and ranking metric: area under the ROC curve (`metric = "ROC"`). `caret` dummy-coded factors automatically; the two SVMs additionally centred and scaled the predictors. The seed (6107) was set before the split, before fold construction, and before every `train` call. **Every complexity hyperparameter** — `lambda`, `nprune`, `min.node.size`, `n.trees`, `interaction.depth`, `shrinkage`, C, σ — **was selected strictly inside its grid**, so the cross-model comparison is apple-to-apple. Three secondary picks land at a family's natural endpoint (GLMNET $\alpha = 1$, MARS `degree=1`, RF `mtry=1`); these are not signs of an under-extended grid but themselves substantive findings — the data prefer the simplest / most-regularised variant within each family.

The candidate set spans interpretable and flexible families: **logistic regression** (`glm`); **penalised logistic regression** (`glmnet`, $\alpha \in [0, 1]$ over 11 values, λ over 30 log-spaced values e^{-8}–e^{-1}); **GAM** with thin-plate smooths on the continuous predictors; **MARS** (`earth`, `degree` 1–3, `nprune` 2–18); **LDA**; **random forest** (`ranger`, `mtry`$\in \{1, \ldots, 8\}$, `min.node.size`$\in \{2, 4, \ldots, 16\}$); **AdaBoost** (`gbm`, `distribution = "adaboost"`, `n.trees`$\in \{1{,}000, 2{,}000, 3{,}000, 4{,}000, 5{,}000\}$, `interaction.depth` 1–4, `shrinkage`$\in \{0.0005, 0.001, 0.003, 0.01, 0.03\}$, `n.minobsinnode = 10`); and two SVMs — **linear** (`svmLinear`, C on a 30-point log grid from e^{-3} to e^{8}) and **radial-basis** (`svmRadialSigma`, C on e^{-5}–e^{5} and σ on e^{-8}–e^{-2}, both with 10–15 grid points). CV tuning curves are in **Figure 2**; selected hyperparameters are in **Table 2**.

4. Results

4.1 Predictive performance — and the boosting / SVM question

CV ROC and held-out test-set AUC, accuracy, sensitivity, specificity, and Brier are in **Table 3** and **Figure 3**; CV ROC distributions are in **Figure 2 (panel e)**. **The answer to the researchers' first question is no — boosting and SVM do not beat the simpler methods.** On 10-fold CV, MARS and GLMNET tie at the top (mean ROC 0.877); AdaBoost is essentially tied (0.876), and SVM linear (0.873), SVM radial, GLM, LDA, and GAM all sit within ≤ 0.01 of the top; **random forest is the only model meaningfully weaker on CV** (0.857). On the test set every AUC falls in the narrow band [0.900, 0.913], a spread of only 0.012 — the 95% bootstrap CIs (Table 3) overlap completely and are each ~0.09 wide. **The test-AUC ranking is therefore essentially**

within sampling noise at $n_{\text{test}} = 200$: RF rises from worst on CV to nominal first on test, while MARS slips from tied-first to last, with neither move exceeding ~1%. The CV ranking is the more reliable summary because it averages over ten folds rather than a single 200-patient draw.

Why don't the flexible methods help? With the four dominant predictors entering the response in nearly additive, threshold-linear form (see §4.3), there is no nonlinear-interaction signal for AdaBoost or the radial SVM to exploit; the extra flexibility chases noise. RF is penalised more — `mtry`=1 with `min.node.size`=10 is the most regularised configuration on its CV curve, and even that loses ~0.02 to the simple methods. Boosting's lower test sensitivity at the 0.5 cutoff (Sens = 0.67) reflects miscalibrated probabilities under the exponential loss, not a shortfall in discrimination — its test AUC is in the same band as the others.

4.2 Predictive risk score

MARS and GLMNET are tied at the top of CV ROC, but only MARS yields a closed-form, additive risk-score expression that a clinician can read off; we therefore use MARS to define $\hat{p}(\text{Severe} \mid x)$. After CV the model reduces to **five terms in four predictors**:

$$\text{logit}(\hat{p}_{\text{Severe}}) = -0.96 + 0.171\, h(\text{age} - 60) + 0.353\, h(\text{bmi} - 28.7) + 0.089\, h(\text{SBP} - 118) - 3.32 \cdot \mathbf{1}[\text{vaccinated}],$$

where $h(u) = \max(u, 0)$, so each continuous risk factor is flat below its breakpoint and rises linearly above it. This is unusually transparent for a "machine-learning" model. The score spans $[0.01, 0.99]$ on the test set. At the default 0.5 cutoff, test sensitivity is 0.76 and specificity 0.92 (test Brier 0.111); clinicians can re-thresh along the ROC curve to trade the two off. Calibration on the held-out test set (**Figure 4**) tracks the diagonal in the low- and mid-risk range; the highest two deciles deviate by about 0.10, but each contains only 20 patients, so the apparent miscalibration is comparable to the per-bin Monte Carlo noise.

4.3 Key predictors and their effects

Three importance measures — MARS term selection, AdaBoost relative influence (**Figure 5a**), and random-forest permutation importance — agree on the top predictors and on their order: **vaccination status, SBP, age, and BMI**, which together account for about 98% of AdaBoost's relative influence. Hypertension is a near-duplicate of SBP (r = 0.80) and falls out of MARS once SBP enters; race, gender, smoking, LDL, and depression each contribute below 1% to AdaBoost and are never selected by MARS.

The partial dependence plots (**Figure 5b**) show *how predicted risk varies with* each predictor in the AdaBoost model (model-implied associations in this observational sample, not causal effects):

- **Vaccination:** by far the strongest single signal. The PDP gap between unvaccinated and vaccinated is 0.44 in $\hat{p}(\text{Severe})$ (0.56 vs 0.13), echoing the empirical-rate gap (62% vs 10%) and the -3.32 MARS vaccine coefficient.
- **Age:** flat below ~58, rising sharply between 58 and 65 to a plateau near 0.40 (consistent with the MARS hinge at 60).
- **BMI:** flat below ~28, then rising steeply between 28 and 33 (PDP climbing from ~0.25 to ~0.40), aligning with the MARS hinge at 28.7.
- **SBP:** rising monotonically from ~120 mmHg, with the steepest segment between 125 and 140. This captures the same signal as binary `hypertension`, which is why hypertension drops out of MARS once SBP is in the model.

These directions are biologically plausible and consistent across MARS hinge coefficients, AdaBoost PDPs, and the GAM smooths (not shown). They should be read as predictive associations rather than causal effects: vaccinated and unvaccinated patients differ systematically in age and comorbidities, so unadjusted comparisons overstate any causal effect of vaccination.

5. Conclusion

For this 1,000-patient sample, **boosting and SVM provide no advantage over simpler models** — the answer to the researchers' first question. On 10-fold CV, MARS and GLMNET tie at the top (mean ROC 0.877), with AdaBoost, the two SVMs, GLM, LDA, and GAM all within ≤ 0.01 of them; only random forest is meaningfully weaker. Test AUCs all sit in [0.900, 0.913] with completely overlapping 95% bootstrap CIs, so the test ranking is dominated by the noise of a 200-patient draw. We select **MARS** as the final model: tied for best on CV and uniquely yielding a five-term, four-predictor closed-form risk-score formula clinicians can compute at the bedside. **The dominant predictors of severe COVID-19 are vaccination status (strongly inversely associated), age, BMI, and systolic blood pressure**, entering the response in a nearly additive, threshold-linear form — which is precisely why the simple methods suffice. The risk score is well-calibrated below the median and re-thresholdable along the ROC curve for any sensitivity/specificity target. **Limitations.** The estimates rest on a single 1,000-patient observational sample without external validation, so the coefficients are predictive associations rather than causal effects; calibration in the top two deciles (Figure 4) is uncertain at $n_{\text{bin}} = 20$ and would need re-checking on a larger or higher-risk validation cohort.

Appendix — Tables and Figures

Table 1: Patient characteristics by COVID-19 severity (training set, n = 800). Continuous variables: mean (SD); categorical variables: n (column %).

	Severe (n = 253)	Not severe (n = 547)
Continuous, mean (SD)		
age	61.3 (4.4)	59.6 (4.4)
bmi	28.0 (2.7)	27.3 (2.6)
SBP	132.8 (7.5)	129.0 (7.9)
height	169.7 (6.4)	170.6 (5.9)
weight	80.5 (6.8)	79.3 (6.9)
LDL	111.9 (20.3)	109.3 (20.0)
depression	7.1 (1.9)	7.1 (2.0)
Categorical, n (column %)		
vaccine		
Yes	46 (18%)	420 (77%)
No	207 (82%)	127 (23%)
gender		
Male	126 (50%)	278 (51%)
Female	127 (50%)	269 (49%)
race		
White	153 (60%)	348 (64%)
Asian	19 (8%)	34 (6%)
Black	48 (19%)	109 (20%)
Hispanic	33 (13%)	56 (10%)
smoking		
Never	153 (60%)	344 (63%)
Former	75 (30%)	160 (29%)
Current	25 (10%)	43 (8%)
hypertension		
Yes	160 (63%)	233 (43%)
No	93 (37%)	314 (57%)
diabetes		
Yes	44 (17%)	75 (14%)
No	209 (83%)	472 (86%)

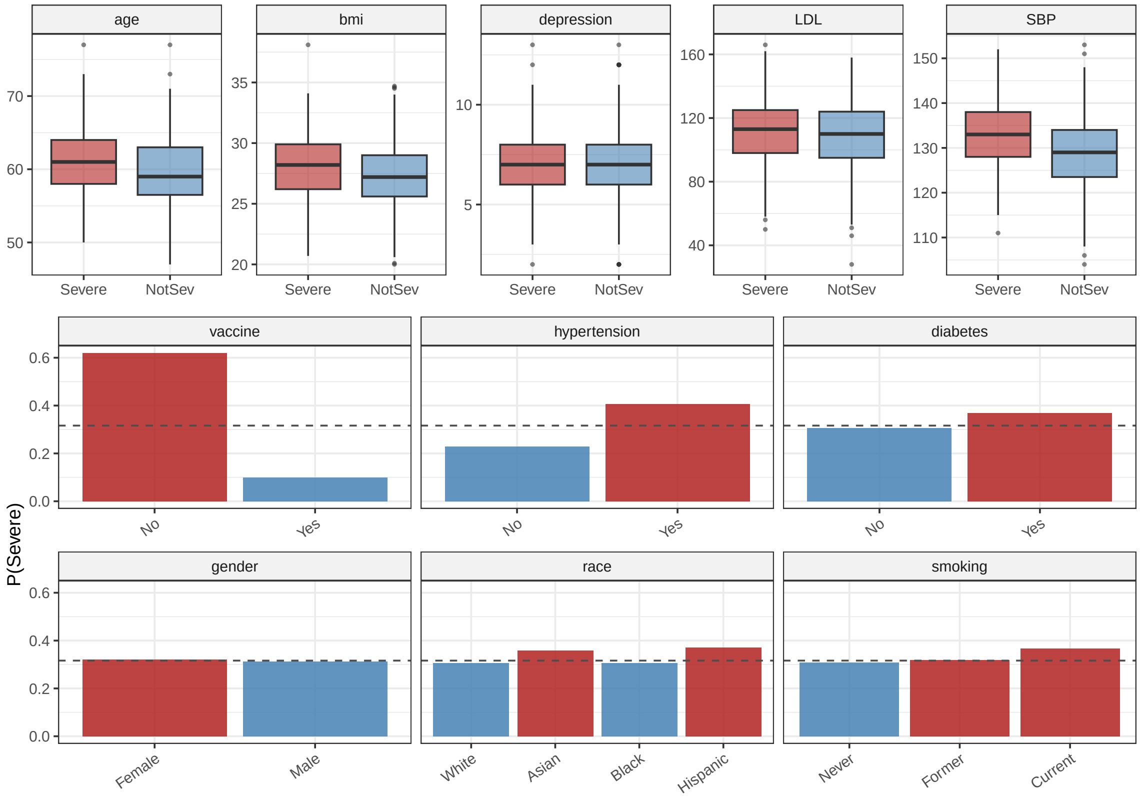

Figure 1: EDA — top: continuous predictors by severity (boxplots, training set, n = 800). Bottom: empirical severity rate by level for categorical predictors; bars above the overall rate of 32% (dashed line) are red, those below are blue.

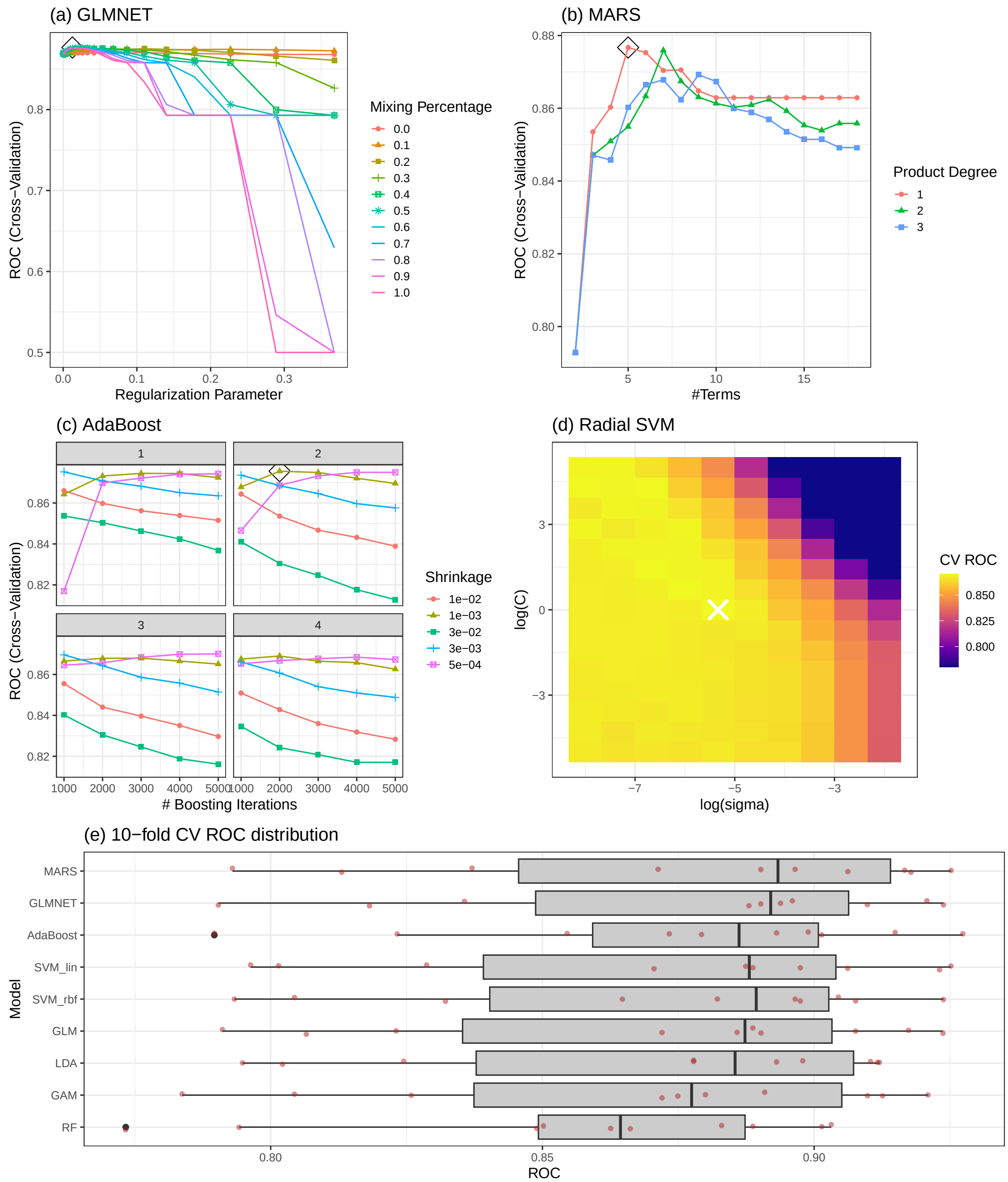

Figure 2: CV tuning curves and the resamples ROC distribution. (a) GLMNET ROC vs $\log\lambda$, coloured by mixing α. (b) MARS ROC vs nprune, coloured by interaction degree. (c) AdaBoost ROC vs n.trees, coloured by interaction depth, faceted by shrinkage. (d) Radial SVM ROC heatmap on $(\log\sigma, \log C)$; the cross marks the CV-selected pair. (e) 10-fold CV ROC distribution by model.

Table 2: Hyper-parameter tuning grids and CV-selected values. Continuous predictors are centred/scaled only for the two SVMs.

Model	Tuning grid (10-fold CV)	Best params
GLM	—	—
GLMNET	alpha: 11 pts in [0,1]; lambda: 30 log-pts in exp(-8..-1)	alpha=1, lambda=0.0125
GAM	default thin-plate smooths on continuous predictors	—
MARS	degree: 1–3; nprune: 2–18	nprune=5, degree=1
LDA	—	—
RF (ranger)	mtry: 1–8; min.node.size: {2,4,...,16}	mtry=1, splitrule=gini, min.node.size=10
AdaBoost (gbm)	n.trees: {1000,...,5000}; depth: 1–4; shrinkage: {5e-4, 1e-3, 3e-3, 0.01, 0.03}	n.trees=2000, interaction.depth=2, shrinkage=0.001, n.minobsinnode=10
SVM linear	C: 30 log-pts in exp(-3..8)	C=447
SVM radial	C: 15 log-pts in exp(-5..5); sigma: 10 log-pts in exp(-8..-2)	sigma=0.00483, C=1

Table 3: Cross-validated and test-set performance for all models. Models sorted by test AUC. 95% CIs are 2,000-replicate stratified bootstrap percentiles (`pROC::ci.auc`). Brier = mean squared error of predicted probabilities; lower is better.

Model	CV ROC (mean)	Test AUC (95% CI)	Test Acc.	Sens.	Spec.	Brier
RF	0.857	0.913 (0.866, 0.952)	0.775	0.259	0.986	0.150
GAM	0.868	0.910 (0.862, 0.948)	0.850	0.724	0.901	0.106
GLM	0.871	0.909 (0.863, 0.950)	0.850	0.724	0.901	0.107
GLMNET	0.877	0.909 (0.859, 0.950)	0.860	0.759	0.901	0.107
LDA	0.870	0.909 (0.860, 0.949)	0.845	0.776	0.873	0.108
SVM_lin	0.873	0.907 (0.859, 0.949)	0.855	0.724	0.908	0.108
SVM_rbf	0.871	0.907 (0.856, 0.949)	0.835	0.741	0.873	0.109
AdaBoost	0.876	0.906 (0.854, 0.948)	0.860	0.672	0.937	0.114
MARS	0.877	0.900 (0.850, 0.945)	0.870	0.759	0.915	0.111

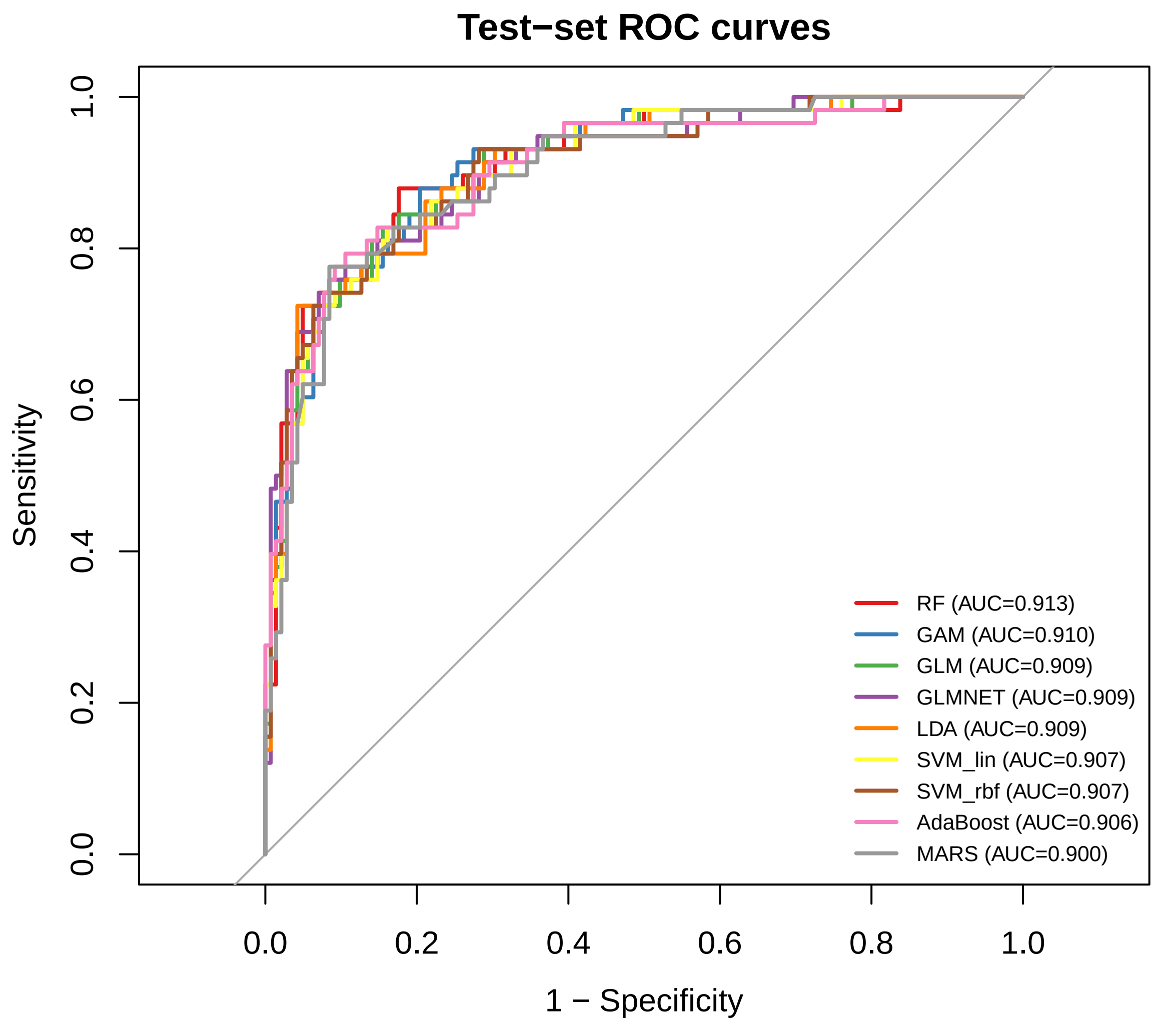

Figure 3: Test-set ROC curves for the nine fitted models with their AUCs.

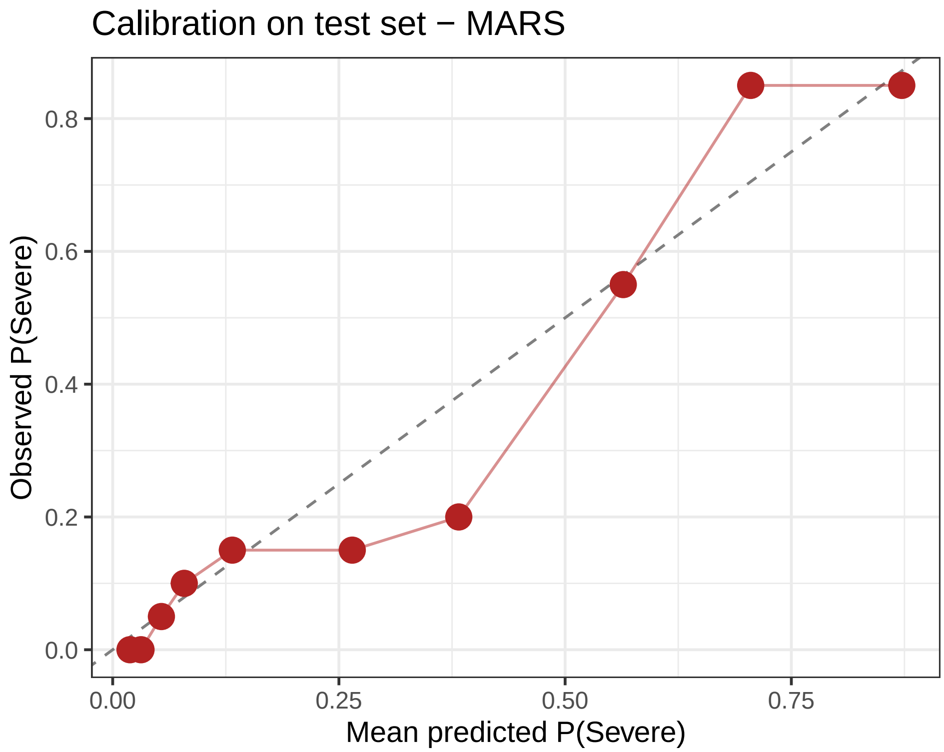

Figure 4: Calibration of the MARS risk score on the held-out test set. Patients are split into 10 equal-size bins by predicted P(Severe); each dot is a bin's mean predicted vs. observed severity rate (point area is bin size). Closer to the dashed diagonal is better.

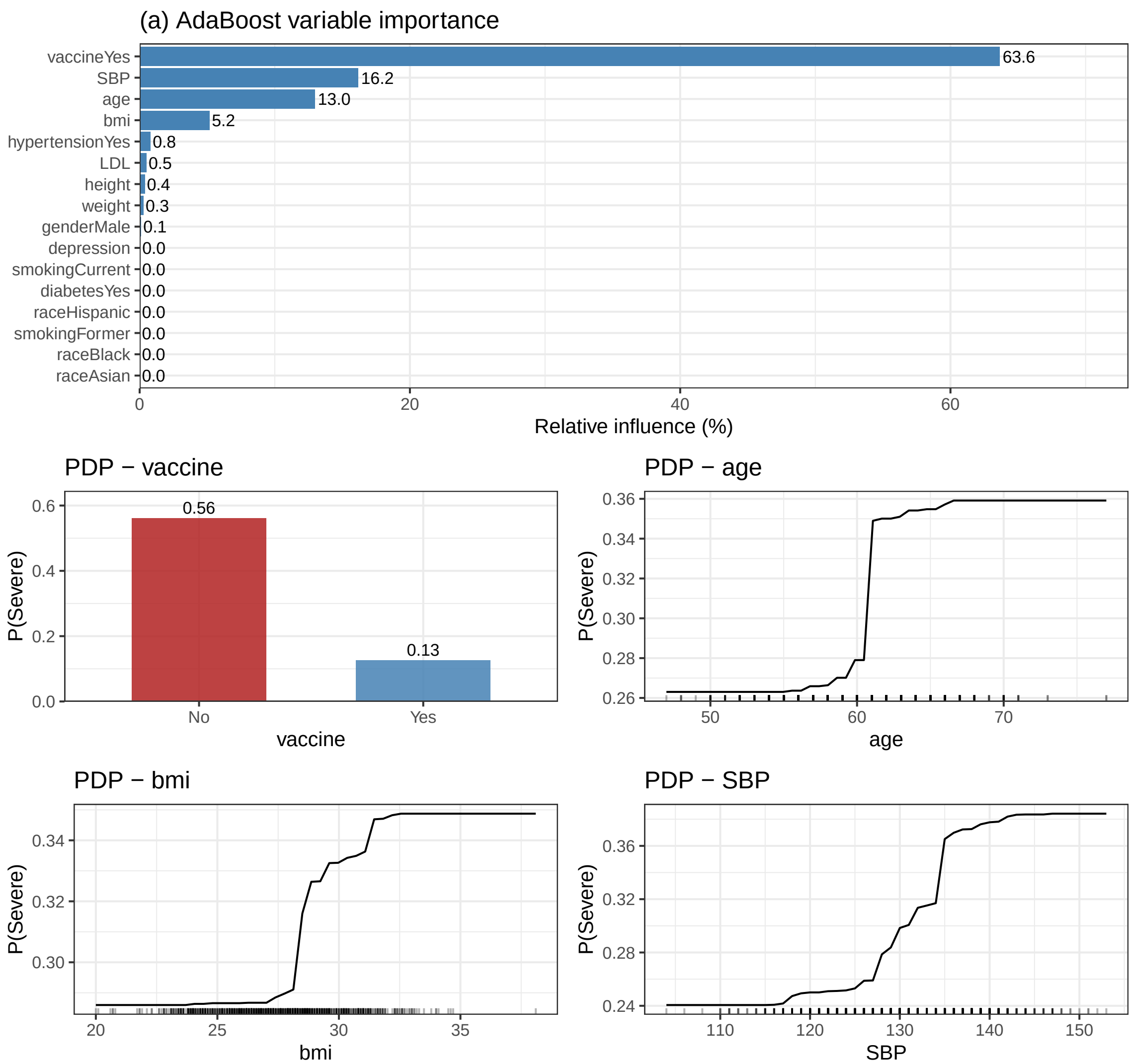

Figure 5: Boosting interpretation. (a) Variable importance (relative influence, %) from the AdaBoost final model — vaccine, SBP, age, and BMI together account for ∼98% of the predictive signal. (b) Partial dependence of the predicted P(Severe) on the four dominant predictors (vaccine, age, BMI, SBP), holding all other predictors at their training distribution.